

4.1.1 Charge

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) electric current as rate of flow of charge;	
$I = \frac{\Delta Q}{\Delta t}$	
(b) the coulomb as the unit of charge	
(c) the elementary charge e equals $1.6 \times 10^{-19} \text{ C}$	Learners will be expected to know that an electron has charge $-e$ and a proton a charge $+e$.
(d) net charge on a particle or an object is quantised and a multiple of e	
(e) current as the movement of electrons in metals and movement of ions in electrolytes	
(f) conventional current and electron flow	
(g) Kirchhoff's first law; conservation of charge.	